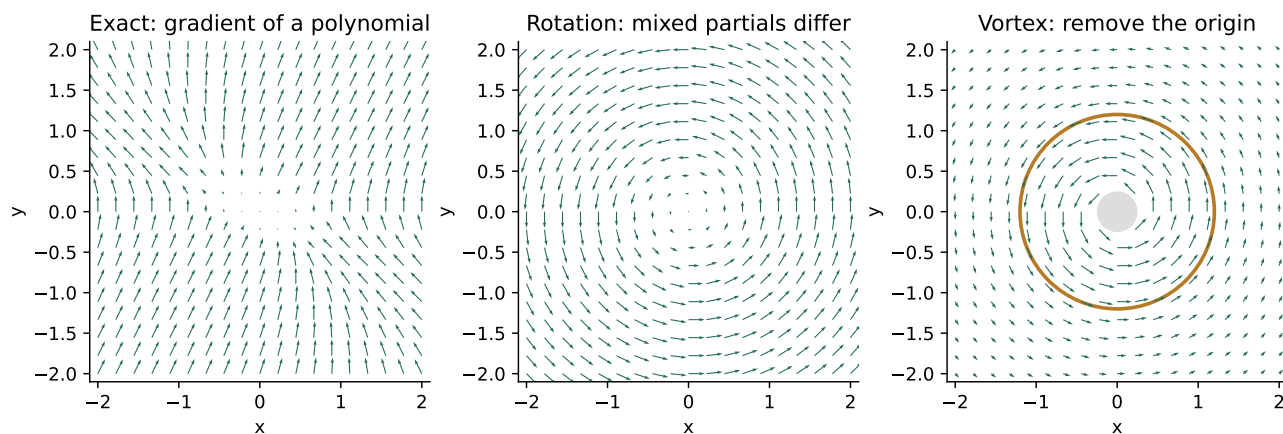


Vector Calculus

An Identity and Proof Atlas

From pictures to tensor algebra

A companion to the interactive Vector Calculus Lab. A comprehensive standard catalogue of Cartesian vector algebra, differential rules, radial fields, integral theorems, and potential tests, with a component or index proof for every numbered statement.



Start with a question. Is this field describing uphill slopes, sources, or circulation? Then track which index is free, which indices are summed, and where the derivative acts.

Contents

1	How to read the proofs	4
2	The algebra behind every curl	4
1.	Delta as the identity tensor	4
2.	Epsilon contraction	4
3.	Cross-product antisymmetry and orthogonality	4
4.	Scalar triple product	5
5.	Vector triple product (BAC-CAB)	5
6.	Lagrange identity and cross-product magnitude	5
7.	Dyad action and transpose	5
3	First derivatives: product, quotient, and chain rules	5
8.	Linearity	5
9.	Gradient of a scalar product	5
10.	Divergence of a scaled vector	6
11.	Curl of a scaled vector	6
12.	Scalar quotient gradient	6
13.	Vector quotient divergence and curl	6
14.	Scalar chain rule	6
15.	Directional product rules	6
16.	Divergence of a cross product	6
17.	Curl of a cross product	7
18.	Gradient of a dot product	7
19.	Convective acceleration identity	7
20.	Divergence of a dyadic field	7
21.	Curl of a gradient cross gradient	7
4	Second derivatives and cancellations	7
22.	Divergence of a gradient	7
23.	Curl of a gradient	8
24.	Divergence of a curl	8
25.	Double curl	8
26.	Laplacian of a scalar product	8
27.	Laplacian of a scaled vector	8
28.	Laplacian chain rule	8
29.	Laplacian of a dot product	8

30. Laplacian of a cross product	9
31. Laplacian commutes with gradient	9
32. Laplacian commutes with divergence and curl	9
33. Triple curl	9
34. Curl of a scalar times a gradient	9
35. Divergence of two crossed gradients	9
36. Curl energy and the antisymmetric Jacobian	9
5 Radial fields: the gravity toolkit	10
37. Position and radius derivatives	10
38. Radial scalar gradient and Hessian	10
39. Radial scalar Laplacian	10
40. Radial divergence and curl	10
41. Power-law Laplacian and inverse-square flux	10
42. Point-source distribution	11
6 From local derivatives to integral theorems	11
43. Gradient theorem	11
44. Divergence theorem	11
45. Stokes theorem	11
46. Green circulation theorem	12
47. Green flux theorem	12
48. Scalar-vector integration by parts	12
49. Green first identity	12
50. Green second identity	12
51. Curl integration by parts	12
52. Vector forms of surface integration	13
53. Green representation formula (three dimensions)	13
7 Potentials and the condition $\mathbf{P} \text{ sub } \mathbf{y} \text{ equals } \mathbf{Q} \text{ sub } \mathbf{x}$	13
54. The planar exactness condition	13
55. Constructing a potential on a rectangle	13
56. Closed-loop criterion and path independence	13
57. Three-dimensional condition and star-shaped construction	14
7.1 Worked example: find a potential rather than guess	14
7.2 Worked example: a failed condition	14
7.3 Worked example: equal mixed partials, no global potential	14

8	Physics and geometry: worked examples	15
8.1	Gravity of a point mass: signs, sources, and work	15
8.2	A uniform massive sphere: what changes inside matter?	15
8.3	Gravity close to Earth: constant-field approximation	16
8.4	Electrostatics: same machinery, different sign	16
8.5	Fluid flow: divergence, vorticity, and acceleration	16
8.6	Harmonic saddles and diffusion	17
8.7	Double curl in electromagnetic waves	17
9	Using the atlas responsibly	17
9.1	Curvilinear coordinates and tensor notation	17
9.2	A short study route	17
9.3	Further reading and conventions	18

1 How to read the proofs

This is a broad standard repertoire, not a claim to enumerate every identity that can be derived. Scalar fields are f, g ; vector fields are $\mathbf{A}, \mathbf{B}, \mathbf{F}$. Work in oriented, orthonormal Cartesian coordinates in Euclidean $\mathbb{R}^3$. Indices i, j, k, ℓ, m run from 1 to 3; repeated indices are summed. A *free* index labels the output component and must match on both sides. Write $\partial_i = \partial/\partial x_i$.

$$(\nabla f)_i = \partial_i f, \quad \nabla \cdot \mathbf{A} = \partial_i A_i, \quad (\nabla \times \mathbf{A})_i = \varepsilon_{ijk} \partial_j A_k, \quad \nabla^2 f = \partial_j \partial_j f.$$

The vector Laplacian means $(\nabla^2 \mathbf{A})_i = \partial_j \partial_j A_i$. The directional derivative is $[(\mathbf{A} \cdot \nabla) \mathbf{B}]_i = A_j \partial_j B_i$. It differentiates $\mathbf{B}$, not $\mathbf{A}$. Use $J_{ij} = \partial_j F_i$ for a Jacobian, $H_{ij} = \partial_i \partial_j f$ for a Hessian, and $(\mathbf{A} \otimes \mathbf{B})_{ij} = A_i B_j$ for a dyad. Our tensor divergence is $(\nabla \cdot \mathbf{T})_i = \partial_j T_{ij}$.

Regularity. First-order identities assume C^1 fields on an open domain; second-order identities assume C^2 ; third-order commutations assume C^3 . Scalars in denominators must be nonzero. Constants are spatially constant. Extra assumptions for integral theorems, potentials, singularities and coordinate systems appear where needed. Mixed derivatives commute only when justified by regularity.

Proof recipe. (1) Write a component. (2) Expand a product or contract two ε symbols. (3) Relabel dummy indices. (4) Recognise the vector expression. Tensor algebra proves the differential rules. Integral theorems additionally need the one-dimensional fundamental theorem of calculus and geometric patching; those ingredients are made explicit.

A sign check. $\varepsilon_{123} = +1$; odd permutations change sign, even ones do not, and any repeated index makes ε zero. A blue dot in the lab means $+z$; a cross means $-z$.

2 The algebra behind every curl

1. Delta as the identity tensor

$$\delta_{ij} a_j = a_i, \quad \delta_{ij} \delta_{jk} = \delta_{ik}, \quad \delta_{ii} = 3.$$

Index proof. By definition $\delta_{ij} = 1$ when $i = j$, otherwise zero. The sum selects the matching component. The matrix product selects $j = i = k$. The trace counts the three coordinates.

2. Epsilon contraction

$$\varepsilon_{ijk} \varepsilon_{ilm} = \delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}.$$

Index proof. If $j = k$ or $\ell = m$, both sides vanish. Otherwise the left side can be nonzero only when the ordered pairs (j, k) and (ℓ, m) contain the same two distinct indices. Its value is $+1$ for the same order and -1 for the reversed order, exactly the right side. In particular $\varepsilon_{ijk} \varepsilon_{ijl} = 2\delta_{kl}$ and $\varepsilon_{ijk} \varepsilon_{ijk} = 6$.

Picture / use. This is the reusable engine for double curls and cross-product identities.

3. Cross-product antisymmetry and orthogonality

$$\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}, \quad \mathbf{A} \cdot (\mathbf{A} \times \mathbf{B}) = 0, \quad \mathbf{A} \times \mathbf{A} = 0.$$

Index proof. $\varepsilon_{ijk} A_j B_k = -\varepsilon_{ikj} A_j B_k = -\varepsilon_{ijk} B_j A_k$. Also $\varepsilon_{ijk} A_i A_j B_k = 0$ because $A_i A_j$ is symmetric in i, j while ε is antisymmetric. The same argument sets $\varepsilon_{ijk} A_j A_k = 0$.

4. Scalar triple product

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}).$$

Index proof. The first expression is $\varepsilon_{ijk}A_iB_jC_k$. Cyclic relabelling $(i, j, k) \mapsto (j, k, i)$ preserves ε , giving the other expressions. Interchanging any two vectors changes the sign. **Picture / use.** The signed volume of a parallelepiped is unchanged by cyclic relabelling.

5. Vector triple product (BAC-CAB)

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}).$$

Index proof. $[\mathbf{A} \times (\mathbf{B} \times \mathbf{C})]_i = \varepsilon_{ijk}\varepsilon_{klm}A_jB_lC_m = (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl})A_jB_lC_m$. Contract to get $B_iA_jC_j - C_iA_jB_j$. Antisymmetry also gives $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{A}(\mathbf{B} \cdot \mathbf{C})$. **Picture / use.** Cross products are not associative. Parentheses matter.

6. Lagrange identity and cross-product magnitude

$$(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) = (\mathbf{A} \cdot \mathbf{C})(\mathbf{B} \cdot \mathbf{D}) - (\mathbf{A} \cdot \mathbf{D})(\mathbf{B} \cdot \mathbf{C}).$$

Index proof. Contract $\varepsilon_{ijk}\varepsilon_{ilm}A_jB_kC_lD_m$ using identity 2. Setting $\mathbf{C} = \mathbf{A}$, $\mathbf{D} = \mathbf{B}$ yields $|\mathbf{A} \times \mathbf{B}|^2 = |\mathbf{A}|^2|\mathbf{B}|^2 - (\mathbf{A} \cdot \mathbf{B})^2$.

7. Dyad action and transpose

$$(\mathbf{A} \otimes \mathbf{B})\mathbf{C} = \mathbf{A}(\mathbf{B} \cdot \mathbf{C}), \quad (\mathbf{A} \otimes \mathbf{B})^T = \mathbf{B} \otimes \mathbf{A}.$$

Index proof. $A_iB_jC_j$ is A_i times the scalar B_jC_j . Transposition exchanges i, j , producing B_iA_j . The trace is $A_iB_i = \mathbf{A} \cdot \mathbf{B}$.

3 First derivatives: product, quotient, and chain rules

8. Linearity

$$D(af + bg) = aDf + bDg, \quad D(a\mathbf{A} + b\mathbf{B}) = aD\mathbf{A} + bD\mathbf{B}.$$

Index proof. Here D is any type-compatible operator among gradient, divergence, curl and Laplacian, and a, b are constants. Each component is a linear combination of partial derivatives. For example $\varepsilon_{ijk}\partial_j(aA_k + bB_k) = a\varepsilon_{ijk}\partial_jA_k + b\varepsilon_{ijk}\partial_jB_k$.

9. Gradient of a scalar product

$$\nabla(fg) = f\nabla g + g\nabla f.$$

Index proof. $\partial_i(fg) = f\partial_i g + g\partial_i f$ for each free index i . **Picture / use.** Both factors can change as you move.

10. Divergence of a scaled vector

$$\nabla \cdot (f\mathbf{A}) = f\nabla \cdot \mathbf{A} + \nabla f \cdot \mathbf{A}.$$

Index proof. $\partial_i(fA_i) = f\partial_i A_i + (\partial_i f)A_i$. **Picture / use.** A changing weight can create sources even when $\nabla \cdot \mathbf{A} = 0$.

11. Curl of a scaled vector

$$\nabla \times (f\mathbf{A}) = f\nabla \times \mathbf{A} + \nabla f \times \mathbf{A}.$$

Index proof. $\varepsilon_{ijk}\partial_j(fA_k) = f\varepsilon_{ijk}\partial_j A_k + \varepsilon_{ijk}(\partial_j f)A_k$. **Picture / use.** Keep the order $\nabla f \times \mathbf{A}$.

12. Scalar quotient gradient

$$\nabla(f/g) = \frac{g\nabla f - f\nabla g}{g^2}.$$

Index proof. $\partial_i(fg^{-1}) = g^{-1}\partial_i f - fg^{-2}\partial_i g$, on a region where $g \neq 0$.

13. Vector quotient divergence and curl

$$\begin{aligned}\nabla \cdot (\mathbf{A}/f) &= \frac{\nabla \cdot \mathbf{A}}{f} - \frac{\nabla f \cdot \mathbf{A}}{f^2}, \\ \nabla \times (\mathbf{A}/f) &= \frac{\nabla \times \mathbf{A}}{f} - \frac{\nabla f \times \mathbf{A}}{f^2}.\end{aligned}$$

Index proof. For $f \neq 0$, expand $\partial_i(f^{-1}A_i)$ and $\varepsilon_{ijk}\partial_j(f^{-1}A_k)$ respectively. Use $\partial_j f^{-1} = -f^{-2}\partial_j f$.

14. Scalar chain rule

$$\nabla h(f) = h'(f)\nabla f.$$

Index proof. $\partial_i h(f) = h'(f)\partial_i f$. More generally $\partial_i h(f, g) = h_f \partial_i f + h_g \partial_i g$.

15. Directional product rules

$$\begin{aligned}(\mathbf{A} \cdot \nabla)(fg) &= f(\mathbf{A} \cdot \nabla)g + g(\mathbf{A} \cdot \nabla)f, \\ (\mathbf{A} \cdot \nabla)(f\mathbf{B}) &= f(\mathbf{A} \cdot \nabla)\mathbf{B} + [(\mathbf{A} \cdot \nabla)f]\mathbf{B}.\end{aligned}$$

Index proof. Multiply $\partial_j(fg)$ or $\partial_j(fB_i)$ by A_j , then apply the ordinary product rule. The operator acts on what follows its parentheses.

16. Divergence of a cross product

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot \nabla \times \mathbf{A} - \mathbf{A} \cdot \nabla \times \mathbf{B}.$$

Index proof. $\partial_i(\varepsilon_{ijk}A_jB_k) = \varepsilon_{ijk}(\partial_i A_j)B_k + \varepsilon_{ijk}A_j\partial_i B_k$. In the first term $\varepsilon_{ijk} = \varepsilon_{kij}$; in the second $\varepsilon_{ijk} = -\varepsilon_{jik}$. Thus the terms are $B_k(\nabla \times \mathbf{A})_k - A_j(\nabla \times \mathbf{B})_j$.

17. Curl of a cross product

$$\begin{aligned}\nabla \times (\mathbf{A} \times \mathbf{B}) &= \mathbf{A} \nabla \cdot \mathbf{B} - \mathbf{B} \nabla \cdot \mathbf{A} \\ &\quad + (\mathbf{B} \cdot \nabla) \mathbf{A} - (\mathbf{A} \cdot \nabla) \mathbf{B}.\end{aligned}$$

Index proof. $\varepsilon_{ijk} \partial_j (\varepsilon_{klm} A_l B_m) = \partial_j (A_i B_j - A_j B_i)$. Expand the two products: $A_i \partial_j B_j - B_i \partial_j A_j + B_j \partial_j A_i - A_j \partial_j B_i$.

18. Gradient of a dot product

$$\begin{aligned}\nabla(\mathbf{A} \cdot \mathbf{B}) &= (\mathbf{A} \cdot \nabla) \mathbf{B} + (\mathbf{B} \cdot \nabla) \mathbf{A} \\ &\quad + \mathbf{A} \times \nabla \times \mathbf{B} + \mathbf{B} \times \nabla \times \mathbf{A}.\end{aligned}$$

Index proof. $[\mathbf{A} \times \nabla \times \mathbf{B}]_i = A_j \partial_i B_j - A_j \partial_j B_i$ by contracting two epsilons. Similarly $[\mathbf{B} \times \nabla \times \mathbf{A}]_i = B_j \partial_i A_j - B_j \partial_j A_i$. Adding the two directional derivatives cancels the negative terms, leaving $\partial_i (A_j B_j)$.

19. Convective acceleration identity

$$(\mathbf{A} \cdot \nabla) \mathbf{A} = \nabla \left(\frac{|\mathbf{A}|^2}{2} \right) - \mathbf{A} \times \nabla \times \mathbf{A}.$$

Index proof. Set $\mathbf{B} = \mathbf{A}$ in the previous identity and divide by two. Directly, $\partial_i (A_j A_j / 2) = A_j \partial_i A_j$, while $[\mathbf{A} \times \nabla \times \mathbf{A}]_i = A_j \partial_i A_j - A_j \partial_j A_i$. **Picture / use.** A steady curved flow can accelerate even if its speed is constant.

20. Divergence of a dyadic field

$$\nabla \cdot (\mathbf{A} \otimes \mathbf{B}) = (\mathbf{B} \cdot \nabla) \mathbf{A} + \mathbf{A} \nabla \cdot \mathbf{B}.$$

Index proof. With our stated convention, $\partial_j (A_i B_j) = B_j \partial_j A_i + A_i \partial_j B_j$. Consequently $\nabla \cdot (\mathbf{v} \otimes \mathbf{v}) = (\mathbf{v} \cdot \nabla) \mathbf{v} + \mathbf{v} \nabla \cdot \mathbf{v}$. **Picture / use.** This connects momentum flux to advective acceleration. Tensor-divergence conventions must be stated.

21. Curl of a gradient cross gradient

$$\begin{aligned}\nabla \times (\nabla f \times \nabla g) &= \nabla f \nabla^2 g - \nabla g \nabla^2 f \\ &\quad + (\nabla g \cdot \nabla) \nabla f - (\nabla f \cdot \nabla) \nabla g.\end{aligned}$$

Index proof. Apply identity 17 to $A_i = \partial_i f$, $B_i = \partial_i g$; alternatively expand $\partial_j ((\partial_i f)(\partial_j g) - (\partial_j f)(\partial_i g))$.

4 Second derivatives and cancellations

22. Divergence of a gradient

$$\nabla \cdot \nabla f = \nabla^2 f.$$

Index proof. $\partial_i (\partial_i f) = \partial_1^2 f + \partial_2^2 f + \partial_3^2 f$. This is the definition of the Cartesian scalar Laplacian. **Picture / use.** Sum the curvatures; a saddle can have zero sum without being flat.

23. Curl of a gradient

$$\nabla \times \nabla f = \mathbf{0}.$$

Index proof. $\varepsilon_{ijk}\partial_j\partial_k f = 0$: the C^2 Hessian is symmetric in j, k , while ε is antisymmetric. Pair each term with the same term after swapping j, k . **Picture / use.** A globally defined smooth height function gives zero circulation around every closed path.

24. Divergence of a curl

$$\nabla \cdot \nabla \times \mathbf{A} = 0.$$

Index proof. $\partial_i \varepsilon_{ijk} \partial_j A_k = \varepsilon_{ijk} \partial_i \partial_j A_k = 0$ by symmetry of the i, j derivatives and antisymmetry of ε . The Cartesian epsilon is constant.

25. Double curl

$$\nabla \times \nabla \times \mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}.$$

Index proof. $\varepsilon_{ijk} \varepsilon_{klm} \partial_j \partial_\ell A_m = (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) \partial_j \partial_\ell A_m = \partial_i \partial_j A_j - \partial_j \partial_j A_i$. **Picture / use.** Divergence-free does not mean double curl is zero; it means double curl equals $-\nabla^2 \mathbf{A}$.

26. Laplacian of a scalar product

$$\nabla^2(fg) = f\nabla^2 g + g\nabla^2 f + 2\nabla f \cdot \nabla g.$$

Index proof. $\partial_i \partial_i(fg) = \partial_i[f\partial_i g + g\partial_i f]$. Expanding gives $f\partial_i^2 g + g\partial_i^2 f + 2(\partial_i f)(\partial_i g)$.

27. Laplacian of a scaled vector

$$\nabla^2(f\mathbf{A}) = f\nabla^2 \mathbf{A} + \mathbf{A}\nabla^2 f + 2(\nabla f \cdot \nabla)\mathbf{A}.$$

Index proof. For each free index i , apply the previous scalar product rule to fA_i : $\partial_j \partial_j(fA_i) = f\partial_j \partial_j A_i + A_i \partial_j \partial_j f + 2(\partial_j f)(\partial_j A_i)$.

28. Laplacian chain rule

$$\nabla^2 h(f) = h'(f)\nabla^2 f + h''(f)|\nabla f|^2.$$

Index proof. $\partial_i \partial_i h(f) = \partial_i[h'(f)\partial_i f] = h'(f)\partial_i^2 f + h''(f)(\partial_i f)^2$. Sum over i .

29. Laplacian of a dot product

$$\nabla^2(\mathbf{A} \cdot \mathbf{B}) = (\nabla^2 \mathbf{A}) \cdot \mathbf{B} + \mathbf{A} \cdot \nabla^2 \mathbf{B} + 2(\partial_j A_i)(\partial_j B_i).$$

Index proof. Expand $\partial_j \partial_j(A_i B_i)$ twice with the product rule. The last term sums both i, j ; it is twice the Frobenius inner product of the two Jacobians, not generally $2(\nabla \cdot \mathbf{A})(\nabla \cdot \mathbf{B})$.

30. Laplacian of a cross product

$$\nabla^2(\mathbf{A} \times \mathbf{B}) = (\nabla^2 \mathbf{A}) \times \mathbf{B} + \mathbf{A} \times \nabla^2 \mathbf{B} + 2 \sum_j (\partial_j \mathbf{A}) \times (\partial_j \mathbf{B}).$$

Index proof. Apply $\partial_\ell \partial_\ell$ to $\varepsilon_{ijk} A_j B_k$. Since epsilon is constant, the three component terms are $\varepsilon_{ijk}[(\nabla^2 A_j)B_k + A_j \nabla^2 B_k + 2(\partial_\ell A_j)(\partial_\ell B_k)]$.

31. Laplacian commutes with gradient

$$\nabla^2 \nabla f = \nabla \nabla^2 f.$$

Index proof. For C^3 f , $\partial_j \partial_j \partial_i f = \partial_i \partial_j \partial_j f$ by commuting partial derivatives. This is a Cartesian statement with constant coefficients.

32. Laplacian commutes with divergence and curl

$$\nabla^2(\nabla \cdot \mathbf{A}) = \nabla \cdot (\nabla^2 \mathbf{A}), \quad \nabla^2(\nabla \times \mathbf{A}) = \nabla \times (\nabla^2 \mathbf{A}).$$

Index proof. For C^3 $\mathbf{A}$, move $\partial_\ell \partial_\ell$ through ∂_i in $\partial_\ell \partial_\ell \partial_i A_i$, or through $\varepsilon_{ijk} \partial_j$ in $\partial_\ell \partial_\ell \varepsilon_{ijk} \partial_j A_k$.

33. Triple curl

$$\nabla \times \nabla \times \nabla \times \mathbf{A} = -\nabla^2(\nabla \times \mathbf{A}).$$

Index proof. Apply double curl to $\nabla \times \mathbf{A}$. Its divergence is zero, so the gradient-of-divergence term vanishes. The result follows for C^3 fields.

34. Curl of a scalar times a gradient

$$\nabla \times (f \nabla g) = \nabla f \times \nabla g.$$

Index proof. $\varepsilon_{ijk} \partial_j (f \partial_k g) = \varepsilon_{ijk} (\partial_j f) (\partial_k g) + f \varepsilon_{ijk} \partial_j \partial_k g$. The second term vanishes by Hessian symmetry.

35. Divergence of two crossed gradients

$$\nabla \cdot (\nabla f \times \nabla g) = 0.$$

Index proof. Expand $\varepsilon_{ijk} \partial_i [(\partial_j f)(\partial_k g)]$. The term with $\partial_i \partial_j f$ vanishes by symmetry in i, j , and the term with $\partial_i \partial_k g$ vanishes by symmetry in i, k .

36. Curl energy and the antisymmetric Jacobian

$$|\nabla \times \mathbf{A}|^2 = (\partial_i A_j)(\partial_i A_j) - (\partial_i A_j)(\partial_j A_i).$$

Index proof. Contract $\varepsilon_{kij} \varepsilon_{k\ell m} (\partial_i A_j)(\partial_\ell A_m)$ to obtain the right side. If $W = (J - J^T)/2$ then $|\nabla \times \mathbf{A}|^2 = 2W_{ij}W_{ij}$. **Picture / use.** Curl records the antisymmetric part of local velocity variation; symmetric strain is different.

5 Radial fields: the gravity toolkit

Here $r = |\mathbf{x}| > 0$, $\hat{\mathbf{r}} = \mathbf{x}/r$, and space has dimension three. These formulas do not include the origin unless a distributional statement explicitly says so.

37. Position and radius derivatives

$$\partial_i x_j = \delta_{ij}, \quad \partial_i r = \frac{x_i}{r}, \quad \partial_i \hat{r}_j = \frac{\delta_{ij} - \hat{r}_i \hat{r}_j}{r}.$$

Index proof. Differentiate $r^2 = x_j x_j$: $2r \partial_i r = 2x_i$. Apply the quotient rule to x_j/r to get $\delta_{ij}/r - x_i x_j/r^3$.

38. Radial scalar gradient and Hessian

$$\begin{aligned} \nabla u(r) &= u'(r) \hat{\mathbf{r}}, \\ \partial_i \partial_j u(r) &= u'' \hat{r}_i \hat{r}_j + \frac{u'}{r} (\delta_{ij} - \hat{r}_i \hat{r}_j). \end{aligned}$$

Index proof. First use $\partial_i u = u' x_i/r$. Differentiate again and insert the unit-radius derivative from the preceding identity. The Hessian has a radial eigenvalue u'' and two tangential eigenvalues u'/r .

39. Radial scalar Laplacian

$$\nabla^2 u(r) = u''(r) + \frac{2}{r} u'(r) = \frac{1}{r^2} \frac{d}{dr} (r^2 u'(r)).$$

Index proof. Trace the radial Hessian: $\hat{r}_i \hat{r}_i = 1$ and $\delta_{ii} = 3$. Thus the trace is $u'' + (3-1)u'/r$. Expand the ordinary derivative to obtain the final form.

40. Radial divergence and curl

$$\nabla \cdot [a(r) \hat{\mathbf{r}}] = a' + \frac{2a}{r}, \quad \nabla \times [a(r) \hat{\mathbf{r}}] = \mathbf{0}.$$

Index proof. Write $F_i = a(r) x_i/r$. Then $\partial_i F_i = a' x_i x_i/r^2 + a(3/r - r^2/r^3)$. For curl, $\partial_j F_k$ is a linear combination of δ_{jk} and $x_j x_k$, both symmetric; contraction with ε_{ijk} vanishes.

41. Power-law Laplacian and inverse-square flux

$$\nabla^2 r^n = n(n+1)r^{n-2}, \quad \nabla^2 \frac{1}{r} = 0 \quad (r > 0), \quad \int_{S_R} \frac{\mathbf{x}}{r^3} \cdot \mathbf{n} dS = 4\pi.$$

Index proof. Insert $u' = nr^{n-1}$ and $u'' = n(n-1)r^{n-2}$ into the radial Laplacian. On the sphere $r = R$, $\mathbf{n} = \mathbf{x}/R$, so $x_i n_i/r^3 = 1/R^2$. Multiply by surface area $4\pi R^2$. **Picture / use.** Zero ordinary divergence away from a source does not mean zero flux around it.

42. Point-source distribution

$$\nabla \cdot \frac{\mathbf{x}}{r^3} = 4\pi\delta^{(3)}(\mathbf{x}), \quad \nabla^2 \frac{1}{r} = -4\pi\delta^{(3)}(\mathbf{x}).$$

Index proof. For a compactly supported smooth test function ψ , distributional divergence gives $-\int (x_i/r^3)\partial_i\psi d^3x$. Excise a ball of radius η . Ordinary divergence is zero outside it, and the outward normal of the excised domain is $-\hat{\mathbf{r}}$. Integration by parts gives $\int_{S_\eta} \psi/\eta^2 dS \rightarrow 4\pi\psi(0)$. The omitted ball contribution tends to zero because r^{-2} is locally integrable in three dimensions. Since $\nabla(1/r) = -\mathbf{x}/r^3$, the Laplacian identity follows. No extra point term appears in this first derivative because the corresponding boundary term scales as η . **Picture / use.** The delta is a source concentrated at a point, not an ordinary finite value assigned at the origin.

6 From local derivatives to integral theorems

Use piecewise smooth boundaries and compatible orientations. Fields extend with the stated regularity to a neighbourhood of the integration region, except in the explicitly excised point-source construction. A boundary component surrounding a hole has the induced opposite orientation.

43. Gradient theorem

$$\int_C \nabla f \cdot d\mathbf{x} = f(\mathbf{x}(b)) - f(\mathbf{x}(a)).$$

Index proof. For a piecewise C^1 parameterisation $x_i(t)$ and $C^1 f$, $\partial_i f(x(t))\dot{x}_i(t) = df(x(t))/dt$. Integrate and apply the one-dimensional fundamental theorem on each segment; endpoint terms telescope.

Picture / use. For a mechanical force $\mathbf{F} = -\nabla U$, work is $U(a) - U(b)$.

44. Divergence theorem

$$\int_{\partial V} F_i n_i dS = \int_V \partial_i F_i dV.$$

Index proof. For a rectangular box, integrate $\partial_i F_i$ first in x_i . The fundamental theorem gives the value of F_i on its two opposite faces with normals $+e_i, -e_i$. Sum i . Subdividing a general bounded region with piecewise smooth (or Lipschitz) boundary cancels all internal face fluxes. The boundary and volume Riemann sums converge to the displayed integrals for $C^1 \mathbf{F}$. **Picture / use.** Only outer faces survive when neighbouring boxes are combined.

45. Stokes theorem

$$\int_{\partial S} F_i dx_i = \int_S \varepsilon_{ijk} \partial_j F_k n_i dS.$$

Index proof. On a smooth parameterised patch $x_i(u, v)$, set $a = F_i x_{i,u}$ and $b = F_i x_{i,v}$. Green's rectangle argument gives $\oint (a du + b dv) = \iint (\partial_u b - \partial_v a) du dv$. The mixed derivatives of x_i cancel, leaving $(\partial_j F_i)(x_{j,u} x_{i,v} - x_{j,v} x_{i,u})$. Epsilon contraction identifies this with $(\nabla \times \mathbf{F})_k \varepsilon_{k\ell m} x_{\ell,u} x_{m,v}$. Patch boundaries cancel on an oriented piecewise smooth surface. Use $C^1 \mathbf{F}$ and C^2 patches. **Picture / use.** The induced boundary direction follows the right-hand rule. A singular point on the surface invalidates the ordinary theorem.

46. Green circulation theorem

$$\oint_{\partial D} (P dx + Q dy) = \iint_D (\partial_x Q - \partial_y P) dx dy.$$

Index proof. On $[a, b] \times [c, d]$, integrate $\partial_x Q$ in x and $-\partial_y P$ in y . These give the right-minus-left and bottom-minus-top edge contributions, exactly the counterclockwise boundary integral. Tile a bounded planar region with piecewise smooth boundary, cancel internal edges and take limits. Requires $P, Q \in C^1$ near $\overline{D}$.

47. Green flux theorem

$$\oint_{\partial D} (P dy - Q dx) = \iint_D (\partial_x P + \partial_y Q) dx dy.$$

Index proof. Apply the previous theorem to the pair $(-Q, P)$. On a positively oriented boundary, the outward normal satisfies $\mathbf{n} ds = (dy, -dx)$, so the boundary form equals $(P, Q) \cdot \mathbf{n} ds$.

48. Scalar-vector integration by parts

$$\int_V f \nabla \cdot \mathbf{A} dV = \int_{\partial V} f \mathbf{A} \cdot \mathbf{n} dS - \int_V \nabla f \cdot \mathbf{A} dV.$$

Index proof. Integrate $\partial_i(f A_i) = f \partial_i A_i + (\partial_i f) A_i$ and use the divergence theorem on $f \mathbf{A}$.

49. Green first identity

$$\int_V (f \nabla^2 g + \nabla f \cdot \nabla g) dV = \int_{\partial V} f \partial_n g dS.$$

Index proof. Set $A_i = \partial_i g$ in the preceding identity. Then $\partial_i A_i = \nabla^2 g$, and $A_i n_i = \partial_n g$. Take $f \in C^1$, $g \in C^2$ near the region.

50. Green second identity

$$\int_V (f \nabla^2 g - g \nabla^2 f) dV = \int_{\partial V} (f \partial_n g - g \partial_n f) dS.$$

Index proof. Write Green's first identity for (f, g) and (g, f) with both C^2 . Subtract; $(\partial_i f)(\partial_i g)$ cancels. This is reciprocity for the Laplacian.

51. Curl integration by parts

$$\int_V [\mathbf{B} \cdot \nabla \times \mathbf{A} - \mathbf{A} \cdot \nabla \times \mathbf{B}] dV = \int_{\partial V} \mathbf{n} \cdot (\mathbf{A} \times \mathbf{B}) dS.$$

Index proof. Integrate the cross-product divergence identity $\partial_i(\varepsilon_{ijk} A_j B_k) = B_i(\nabla \times \mathbf{A})_i - A_i(\nabla \times \mathbf{B})_i$ and apply the divergence theorem. The boundary sign is fixed by $\mathbf{n} \cdot (\mathbf{A} \times \mathbf{B})$.

52. Vector forms of surface integration

$$\int_V \nabla f dV = \int_{\partial V} f \mathbf{n} dS, \quad \int_V \nabla \times \mathbf{A} dV = \int_{\partial V} \mathbf{n} \times \mathbf{A} dS.$$

Index proof. For the first, apply the divergence theorem componentwise to $F_j = f\delta_{ij}$ with fixed i . For the second use $F_j = \varepsilon_{ijk}A_k$ with fixed i . Their divergences are $\partial_i f$ and $\varepsilon_{ijk}\partial_j A_k$ respectively.

53. Green representation formula (three dimensions)

$$f(\mathbf{x}) = \int_{\partial V} [\Gamma \partial_n f - f \partial_n \Gamma] dS - \int_V \Gamma \nabla^2 f dV,$$

$$\Gamma(\mathbf{x}, \mathbf{y}) = \frac{1}{4\pi|\mathbf{x} - \mathbf{y}|}, \quad \mathbf{x} \in V.$$

Index proof. Assume $f \in C^2$ near a bounded smooth region and $\mathbf{x}$ strictly inside. In the $\mathbf{y}$ variable, $\nabla_{\mathbf{y}}^2 \Gamma = -\delta^{(3)}(\mathbf{y} - \mathbf{x})$. Apply Green's second identity with $g = \Gamma$, excising a ball about $\mathbf{x}$ and taking its radius to zero (as in identity 42). The volume term becomes $-f(\mathbf{x}) - \int_V \Gamma \nabla^2 f$. Rearranging the outer boundary term $\int_{\partial V} (f \partial_n \Gamma - \Gamma \partial_n f)$ yields the formula. Boundary points require a separate limiting formula.

7 Potentials and the condition $\mathbf{P} \text{ sub } \mathbf{y} \text{ equals } \mathbf{Q} \text{ sub } \mathbf{x}$

54. The planar exactness condition

$$\mathbf{F} = (P, Q) = \nabla \phi \implies \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

Index proof. $P = \partial_x \phi$, $Q = \partial_y \phi$. For C^2 ϕ , $\partial_y P = \partial_y \partial_x \phi = \partial_x \partial_y \phi = \partial_x Q$. In tensor language the Jacobian $J_{ij} = \partial_j F_i$ is symmetric. Conversely, for C^1 $\mathbf{F}$ on an open simply connected planar domain, this condition is sufficient for a single-valued global potential, by the path-independence argument below.

Picture / use. These are *partial* derivatives because P, Q depend on both variables. Compare the off-diagonal derivatives, not P_x and Q_y .

55. Constructing a potential on a rectangle

$$\phi(x, y) = \int_{x_0}^x P(s, y_0) ds + \int_{y_0}^y Q(x, t) dt.$$

Index proof. Assume the rectangle lies in a C^1 domain and $P_y = Q_x$. Clearly $\phi_y = Q(x, y)$. Also $\phi_x = P(x, y_0) + \int_{y_0}^y Q_x(x, t) dt = P(x, y_0) + \int_{y_0}^y P_y(x, t) dt = P(x, y)$. Thus the mixed-partial test is constructively sufficient locally.

56. Closed-loop criterion and path independence

$$\mathbf{F} = \nabla \phi \iff \oint_C \mathbf{F} \cdot d\mathbf{x} = 0 \text{ for every closed path } C.$$

Index proof. On an open connected domain with C^1 $\mathbf{F}$, the forward implication is the gradient theorem. For the reverse, choose a base point and define $\phi(\mathbf{x}) = \int_{\mathbf{x}_0}^{\mathbf{x}} \mathbf{F} \cdot d\boldsymbol{\ell}$. Two paths differ by a closed path, so the value is independent of path. Appending a short coordinate segment gives $\partial_i \phi = F_i$.

Curl-free fields on a simply connected domain have zero loop integral: contract each loop to a point and use Stokes on smooth homotopy patches, where the normal curl is zero. General piecewise smooth loops follow by approximation. Simple connectivity is sufficient, not necessary for a particular field. **Picture / use.** A hole permits a failure but does not force it. All closed-loop integrals being zero is the exact global test.

57. Three-dimensional condition and star-shaped construction

$$\partial_i F_j = \partial_j F_i \iff \nabla \times \mathbf{F} = 0,$$

$$\phi(\mathbf{x}) = \int_0^1 x_i F_i(t\mathbf{x}) dt \quad \text{on a domain star-shaped about } 0.$$

Index proof. Epsilon contraction shows $\nabla \times \mathbf{F} = 0$ is equivalent to the antisymmetric part of $\partial_i F_j$ vanishing. For $C^1 \mathbf{F}$, differentiate the integral: $\partial_k \phi = \int_0^1 [F_k(t\mathbf{x}) + tx_i \partial_k F_i(t\mathbf{x})] dt$. Symmetry turns $\partial_k F_i$ into $\partial_i F_k$, making the integrand $d[tF_k(t\mathbf{x})]/dt$. Hence $\partial_k \phi = F_k(\mathbf{x})$. Translate the base point for any other star centre. In (P, Q, R) notation, the conditions are $P_y = Q_x$, $P_z = R_x$, $Q_z = R_y$.

7.1 Worked example: find a potential rather than guess

Let $P = 2xy + y^2$ and $Q = x^2 + 2xy + 3y^2$. Then $P_y = 2x + 2y = Q_x$ on all of $\mathbb{R}^2$. Integrate $\phi_x = P$:

$$\phi = x^2 y + xy^2 + h(y), \quad \phi_y = x^2 + 2xy + h'(y) = Q.$$

Therefore $h' = 3y^2$ and $\phi = x^2 y + xy^2 + y^3 + C$. From $(0, 0)$ to $(1, 1)$, every path has $\int \mathbf{F} \cdot d\mathbf{x} = 3$. On the horizontal-then-vertical path, the first segment contributes zero and the second gives $\int_0^1 (1+2y+3y^2) dy = 3$. On the diagonal $x = y = t$, $P = 3t^2$, $Q = 6t^2$, so $\int_0^1 9t^2 dt = 3$.

7.2 Worked example: a failed condition

For $P = -\omega y$, $Q = \omega x$, $P_y = -\omega$ and $Q_x = \omega$. Unless $\omega = 0$, they differ, the curl is 2ω , and a counterclockwise radius- R circle has circulation $2\pi\omega R^2$. From $(0, 0)$ to $(1, 1)$, horizontal-then-vertical work is ω ; vertical-then-horizontal work is $-\omega$. The difference 2ω equals the curl integral over the unit square.

7.3 Worked example: equal mixed partials, no global potential

On $\mathbb{R}^2 \setminus \{0\}$, take $P = -by/(x^2 + y^2)$ and $Q = bx/(x^2 + y^2)$. With $s = x^2 + y^2$,

$$P_y = b(y^2 - x^2)/s^2 = Q_x.$$

Nevertheless $\mathbf{x}(\theta) = R(\cos \theta, \sin \theta)$ gives

$$\mathbf{F} \cdot d\mathbf{x} = b d\theta, \quad \oint \mathbf{F} \cdot d\mathbf{x} = 2\pi b.$$

For $b \neq 0$ there is no single-valued global scalar potential. On a slit plane a branch of $\phi = b \arg(x + iy)$ works; its derivative gives the field, but its angle cannot be continuous and single valued on the whole punctured plane. Stokes cannot span the loop with a disk containing the undefined origin. On an annulus the outer and oppositely oriented inner boundary circulations cancel, agreeing with zero ordinary curl on the annulus.

Dimension matters. The punctured plane is not simply connected. In contrast, $\mathbb{R}^3 \setminus \{0\}$ is simply connected: a loop can move around a missing point in the third dimension. Point-mass gravity has the global potential $-GM/r$ there. A missing *line* in three dimensions can reproduce the planar vortex obstruction.

8 Physics and geometry: worked examples

8.1 Gravity of a point mass: signs, sources, and work

Newtonian potential per unit test mass and acceleration are

$$\Phi = -\frac{GM}{r}, \quad g_i = -\partial_i \Phi = -GM \frac{x_i}{r^3}.$$

Using the radial gradient identity gives the inward direction immediately. Direct differentiation away from $r = 0$ gives

$$\partial_j g_i = -GM \left(\frac{\delta_{ij}}{r^3} - \frac{3x_i x_j}{r^5} \right).$$

This Jacobian is symmetric, so $\nabla \times \mathbf{g} = 0$. Its trace is $-GM(3/r^3 - 3r^2/r^5) = 0$, so vacuum has zero ordinary divergence. Yet sphere flux is $-4\pi GM$. The point-source identity resolves the difference:

$$\nabla \cdot \mathbf{g} = -4\pi GM \delta^{(3)}(\mathbf{x}), \quad \nabla^2 \Phi = 4\pi GM \delta^{(3)}(\mathbf{x}).$$

For a smooth mass density, linear superposition gives $\nabla^2 \Phi = 4\pi G\rho$ and $\nabla \cdot \mathbf{g} = -4\pi G\rho$. Negative divergence describes inward flux to positive mass. A test mass m moving from radius r_a to r_b receives work $m[\Phi(r_a) - \Phi(r_b)] = GMm(1/r_b - 1/r_a)$.

Planar mixed-partial check. In the equatorial plane let $P = -GMx/s^{3/2}$, $Q = -GMy/s^{3/2}$ with $s = x^2 + y^2 > 0$. Then $P_y = 3GMxy/s^{5/2} = Q_x$. The potential is explicitly $-GM/\sqrt{s}$. Zero planar curl agrees with this. Do not compute a two-dimensional divergence and interpret it as the three-dimensional vacuum divergence: the missing $\partial_z g_z$ is essential.

8.2 A uniform massive sphere: what changes inside matter?

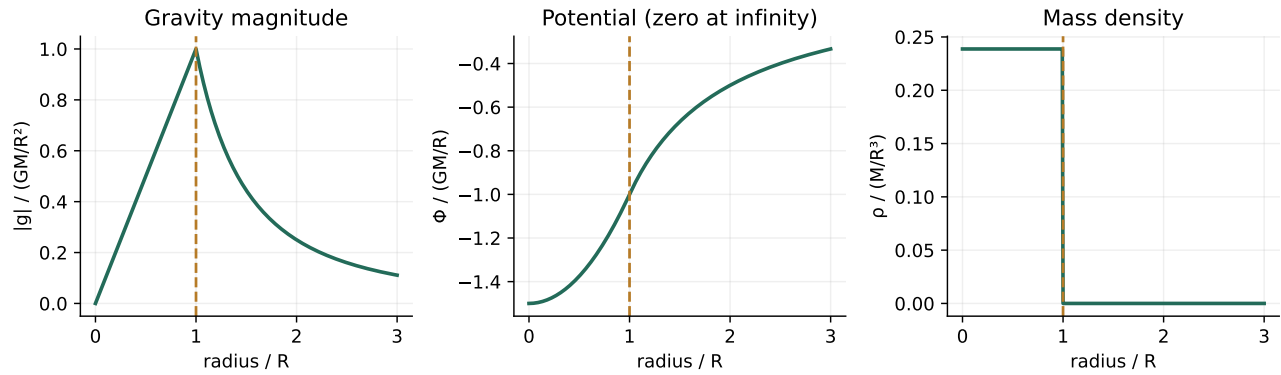
For radius R , total mass M , and density $\rho = 3M/(4\pi R^3)$ inside, spherical symmetry and the divergence theorem give

$$4\pi r^2 g_r = -4\pi GM_{\text{enclosed}}(r), \quad g_r = -\frac{GM_{\text{enclosed}}(r)}{r^2}.$$

Integrating the density gives $M_{\text{enclosed}} = M(r/R)^3$ for $r < R$ and M for $r \geq R$. Hence

$$\mathbf{g} = \begin{cases} -GM\mathbf{x}/R^3, & r < R, \\ -GM\mathbf{x}/r^3, & r > R, \end{cases} \quad \Phi = \begin{cases} -\frac{GM}{2R^3}(3R^2 - r^2), & r \leq R, \\ -GM/r, & r \geq R. \end{cases}$$

Integrate $\Phi' = -g_r$, impose zero at infinity, and match at R to obtain this potential. Inside, $\partial_i g_i = -3GM/R^3 = -4\pi G\rho$ and $\varepsilon_{ijk} \partial_j g_k = 0$; outside use the point-mass calculation. Both potential and normal field are continuous at R . Second derivatives jump, so the classical piecewise equations hold away from the boundary, and the Poisson equation holds weakly across it. There is no surface delta because the normal field has no jump.



The acceleration magnitude rises linearly from zero at the centre, reaches its maximum at the surface, then decays as r^{-2} . The potential is a smooth bowl inside and negative everywhere. The dashed line is the surface; these plots use $GM = R = 1$.

8.3 Gravity close to Earth: constant-field approximation

With upward height z , use $\Phi = g_0 z + C$ and $\mathbf{g} = (0, 0, -g_0)$ on a small patch where g_0 is nearly constant. Every spatial derivative of $\mathbf{g}$ is zero, so both divergence and curl vanish in this approximation. It is not a global model of Earth or its mass distribution. Lifting mass m through height h changes potential energy by $mg_0 h$.

8.4 Electrostatics: same machinery, different sign

For a positive point charge q , $V = q/(4\pi\epsilon_0 r)$ and $\mathbf{E} = -\nabla V = q\mathbf{x}/(4\pi\epsilon_0 r^3)$. The radial identities give $\nabla \times \mathbf{E} = 0$ away from the source and $\nabla \cdot \mathbf{E} = q\delta^{(3)}(\mathbf{x})/\epsilon_0$ distributionally. For a smooth density, $\nabla^2 V = -\rho_e/\epsilon_0$. Gravity draws arrows inward to positive mass; a positive electric charge sends arrows outward. Electrostatic curl is zero, whereas time-varying electromagnetic fields can have $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$.

The smooth cloud used in the sliders. To avoid a singularity in the electric-source experiment, use $V = q/(4\pi\sqrt{r^2 + s^2})$ with $s > 0$ and $\epsilon_0 = 1$. Differentiation gives $E_i = qx_i/[4\pi(r^2 + s^2)^{3/2}]$. Tracing its Jacobian gives $\rho_e = 3qs^2/[4\pi(r^2 + s^2)^{5/2}]$. The outward flux through radius r is $qr^3/(r^2 + s^2)^{3/2}$, which tends to q at infinity. Thus the cloud has total charge q , not a point source at its centre.

8.5 Fluid flow: divergence, vorticity, and acceleration

Rigid rotation $\mathbf{v} = (-\omega y, \omega x, 0)$ has $\partial_i v_i = 0$ but $(\nabla \times \mathbf{v})_z = \partial_x v_y - \partial_y v_x = 2\omega$. Its convective acceleration is

$$(\mathbf{v} \cdot \nabla)\mathbf{v} = (-\omega^2 x, -\omega^2 y, 0).$$

The kinetic-energy gradient is $(\omega^2 x, \omega^2 y, 0)$, while $\mathbf{v} \times \nabla \times \mathbf{v} = (2\omega^2 x, 2\omega^2 y, 0)$; their difference matches the inward acceleration. Here no compression is needed for rotation.

Straight shear $\mathbf{v} = (\gamma y, 0, 0)$ also has zero divergence but curl $(0, 0, -\gamma)$, even though its streamlines are straight. Its convective acceleration is zero because $v_j \partial_j v_i = \gamma y \partial_x v_i = 0$. The gradient $\nabla(|\mathbf{v}|^2/2) = (0, \gamma^2 y, 0)$ equals $\mathbf{v} \times \nabla \times \mathbf{v}$ and cancels. Curl measures local circulation, not simply whether the drawn streamlines bend.

8.6 Harmonic saddles and diffusion

For temperature $T = ax^2 + by^2 + cz^2$, $\partial_i \partial_i T = 2(a + b + c)$. A saddle $x^2 - y^2$ is harmonic even though its gradient $(2x, -2y, 0)$ is nonzero almost everywhere. In homogeneous diffusion $\partial_t T = \kappa \nabla^2 T$ with constant κ , summed curvature controls the instantaneous local change. The bowl $x^2 + y^2$ has $\nabla^2 T = 4$; the saddle has zero local diffusion rate for that ideal field. Initial and boundary conditions are still required to solve an evolution problem.

8.7 Double curl in electromagnetic waves

In a homogeneous, source-free region with constant μ, ε ,

$$\nabla \times \mathbf{E} = -\partial_t \mathbf{B}, \quad \nabla \times \mathbf{B} = \mu\varepsilon \partial_t \mathbf{E}, \quad \nabla \cdot \mathbf{E} = 0.$$

Taking curl of the first equation and commuting smooth space/time derivatives gives $\nabla \times \nabla \times \mathbf{E} = -\mu\varepsilon \partial_t^2 \mathbf{E}$. The double-curl identity reduces the left side to $-\nabla^2 \mathbf{E}$, yielding $\nabla^2 \mathbf{E} = \mu\varepsilon \partial_t^2 \mathbf{E}$. Plane waves therefore have speed $1/\sqrt{\mu\varepsilon}$. This reduction relies on the stated constitutive and source assumptions.

9 Using the atlas responsibly

9.1 Curvilinear coordinates and tensor notation

These proofs used a fixed Cartesian basis. In cylindrical or spherical coordinates, basis vectors vary with position. Simply replacing Cartesian partial derivatives with derivatives of physical spherical components gives wrong answers. In arbitrary coordinates on flat Euclidean space, use the metric g_{ij} and its covariant derivative ∇_i , with appropriate index raising: $\text{div } \mathbf{A} = \nabla_i A^i$, $(\text{grad } f)^i = g^{ij} \partial_j f$ and $\Delta f = \nabla_i \nabla^i f$. The volume Levi-Civita *tensor* replaces the constant symbol. Flat-space identities remain valid when all objects are transformed consistently. On curved manifolds derivative commutations can introduce curvature terms, so the Cartesian double-curl proof cannot be transplanted blindly.

9.2 A short study route

First master identities 1–5 and the meanings of free and dummy indices. Next prove the three scaled-field rules without looking. Then use epsilon contraction to derive double curl and curl of a cross product. Finally test potentials: calculate the off-diagonal derivatives, inspect the domain, construct a potential when possible, and verify its gradient. Use the gravity examples to practise the difference between a smooth vacuum calculation and a source represented distributionally.

9.3 Further reading and conventions

The proofs and worked examples in this atlas are original expositions of standard identities. The following primary educational sources provide additional background:

- OpenStax, *Calculus Volume 3*, sections 6.3–6.7: conservative fields, Green, Stokes and divergence theorems.
<https://openstax.org/books/calculus-volume-3/pages/6-3-conservative-vector-fields>
- MIT OpenCourseWare, *18.02SC Multivariable Calculus*, vector fields and line integrals.
<https://ocw.mit.edu/courses/18-02sc-multivariable-calculus-fall-2010/>
- Richard Fitzpatrick, University of Texas, *Gravitational Potential of Uniform Spheroid*, for Poisson-equation methods in gravity.
<https://farside.ph.utexas.edu/teaching/355/Surveyhtml/node61.html>

Rebuild. Run `python scripts/build_atlas.py` from the repository. Python needs NumPy and Matplotlib; a TeX installation needs pdfLaTeX and the packages listed in the source preamble. The script regenerates the two exact figures, the LaTeX source and this PDF. No network access is required during the build.